

The background features a pattern of vertical bars in two shades of blue. Some bars are a medium blue, while others are a lighter, pale blue. They are arranged in groups of five, with some groups appearing more prominent than others, creating a sense of depth and movement.

# **DataBricks Runtime** **UPGRADED**

AusNet DnA Platform

# Solution Approach



## Runtime Modernisation

Spark 3.2 → 4.0, Python 3.8 → 3.12,  
Scala 2.12 → 2.13 with full API  
compatibility validation



## Blue-Green Deploy

Parallel runtime clusters with instant  
rollback capability — zero downtime,  
zero data loss



## 6-Category Remediation

32 notebooks fixed across API changes,  
connector migration, syntax and config  
updates

## Scope of Work

### 1 Assessment

225 production notebooks scanned  
for breaking changes across Spark,  
Python and Scala

### 2 Remediation

32 notebooks updated — EventHub  
→ Kafka migration, deprecated API  
replacement, config alignment

### 3 Validation

Full regression across Dev, Test and  
Prod with 100% pass rate and zero  
Sev 1 - 4 defects

# Six Categories of Change

1

## API Breaking Changes

Deprecated Spark 3.x APIs replaced with Spark 4.0 equivalents across DataFrame and SQL operations

2

## Connector Migration

EventHub connector deprecated — 11 notebooks migrated to Kafka-based streaming API

3

## Syntax Updates

Python 3.12 syntax enforcement — f-string changes, type hint updates and deprecation removals

4

## Config Alignment

Spark config keys renamed or removed in 4.0 — cluster policies and job configs updated

5

## Scala Cross-Build

Scala 2.12 → 2.13 migration with binary-incompatible library rebuilds and import fixes

6

## Security Patches

Credential handling and secret scope access patterns updated to current Databricks standards

# Upgrade at a Glance

100%

TEST PASS RATE

225

NOTEBOOKS VALIDATED

0

OUTSTANDING DEFECTS

## What We Delivered

- Spark 3.2 → 4.0, Python 3.8 → 3.12, Scala 2.12 → 2.13
- 32 notebooks remediated across 6 change categories
- Blue-green deployment with proven rollback capability
- Zero data loss, zero business disruption

## Business Outcomes

- ✓ Modern LTS runtime with extended support
- ✓ Photon-enabled clusters for faster queries
- ✓ Security and compliance posture restored

## Challenges Overcome

- ⚠ Environment sync issues between Dev, Test and Prod delayed code promotion by 7 days
- ⚠ Oracle and SQL Server connectivity required IT escalation across environments
- ⚠ EventHub connector deprecated — migrated 11 notebooks to Kafka API
- ⚠ 5 pre-existing production issues identified and isolated from project scope